UNIT-4

Choose the correct answer:

1. b) in case of Sc, 3d orbitals are partially filled but in Zn these are completely filled
2. a) Cr

   $Cr \Rightarrow [Ar]3d^5 4s^1$
3. a) Ti
4. c) Ni^{2+}
5. a) 5.92BM

   $Mn^{2+} \Rightarrow 3d^5$ contains 5 unpaired electrons

   n=5 ; $\therefore = \sqrt{n(n+2)}BM$

   $\therefore = \sqrt{5(5+2)} = \sqrt{35} = 5.92BM$
6. c) their ability to adopt variable oxidation states
7. a) $VO_2^+ < Cr_2O_7^{2-} < MnO_4^-$

   $\overset{+5}{V}O_2^+ < \overset{+6}{Cr}_2O_7^{2-} < \overset{+7}{Mn}O_4^-$ greater the oxidation state, higher is the oxidising power
8. b) Carbon dioxide

   $5(COO)_2^{2-} + 2MnO_4^- + 16H^+ \longrightarrow 2Mn^{2+} + 10CO_2 \uparrow + 8H_2O$
9. b) $Na_2Cr_2O_7$ is preferred over $K_2Cr_2O_7$ in volumetric analysis
10. b) Mn^{2+}

    $MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$
11. c) 3

    $K_2Cr_2O_7 + 6KI + 7H_2SO_4 \longrightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$
12. c) 0.6

    $\overset{7+}{Mn}O_4^- + \overset{+2}{Fe}\overset{+3}{C}_2O_4 \longrightarrow Mn^{2+} + Fe^{3+} + 2\overset{+4}{C}O_2$

    $5e^-$ acception $3e^-$ release

    5 moles of $FeC_2O_4 \equiv$ 3 moles of $KMnO_4$

    1 mole of $FeC_2O_4 \equiv \left(\frac{3}{5}\right)$ moles of $KMnO_4$

    1 mole of $FeC_2O_4 \equiv$ 0.6 moles of $KMnO_4$
13. c) All the lanthanons are much more reactive than aluminium.

    As we move from La to Lu , their metallic behaviour because almost similar to that of aluminium.
14. b) Yb^{2+}

    Yb^{2+} - $4f^{14}$ –no unpaired electrons - diamagnetic

Answers.indd 236 22-12-2022 19:40:02